Phase One | Gradient descent

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

Goal: predict hourly rental counts using a linear model trained by our own gradient descent. The supplied training file is the only labeled source; no external data is used.

IDs are sorted as strings and SHA-256 hashed modulo 100000, producing seed 29578. One train_test_split(test_size=0.20) produces 8708 training and 2178 validation rows, reused throughout.

Represent hour, month, weekday, season and weather as reference-category indicators; retain binary calendar and continuous weather features. Derive day-of-month from the date. Fit all means/scales on training only. Pool weather codes 3 and 4 because severe weather is absent from this training portion.

Minimize half-MSE using our full-batch gradient. Compare rates 0.1/L, 0.5/L and 1/L with an excessive 2.2/L. Stop at relative change below 1e-10 for 20 updates, with a 12,000-step cap. L is the largest training-Hessian eigenvalue; rates below 2/L are stable. Finite differences check the derivative; independent least squares checks the achieved training objective. An elapsed-date feature was removed after near redundancy caused false convergence.

Selected rate 0.02763121, 9938 updates. Validation R2 0.69795, RMSE 99.46491 rentals/hour. The saved coefficients and settings initialize Phase Two.

Outcome: categorical hours capture daily peaks, but high-demand underprediction and negative raw predictions remain. A converged model can still perform poorly on future dates.

Bonus (+5%): define e_i=x_i^T w-y_i and a_i=3 if e_i<0, else 1. Use J=sum(a_i e_i^2)/(2m): equal-sized underprediction costs 3 times overprediction. Its prediction derivative is 3e_i below zero and e_i above zero; both are zero at zero. Thus gradient J=X^T(a*e)/m. Recompute a every update. This explicitly uses weighted squared cost, not linear cost.

Our own bonus GD starts from the MSE weights, with unchanged rows/features/scaling and rate 1/(3L)=0.09210404. It reaches training gradient max 1e-05 after 4338 updates (tolerance 1e-5; cap 50,000). Finite differences check all coefficients. No validation stopping or library optimizer is used.

Bonus validation: asymmetric half-MSE 10750.60 -> 8595.38 (20.05% lower); mean prediction 194.19 -> 236.41. Underprediction frequency 44.67% -> 30.76%; RMSE 99.46 -> 108.77. The notebook plots the shift. This cost/RMSE tradeoff is a side experiment; Phase Two still consumes the original MSE weights.

Source: supplied CSV files and executed bike_sharing_demand.ipynb

Phase Two | Polynomial regression

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

Consume Phase One weights and its exact split. Compare selected individual weather powers at degrees 2 and 3 against degree 1; do not expand every indicator or create every interaction.

Expand temperature, humidity and wind speed. Keep feels-like temperature linear because its training correlation with temperature is about 0.986. This is an initial feature choice, not proof that other effects are useless.

Powers are built from training-standardized features; added terms receive training-only centering/scaling. Preserve existing linear columns and initialize added coefficients to zero. Assertions verify that the expanded starting predictions exactly reproduce Phase One.

Train with the same gradient-descent function. Expansion changes the stable rate range, so compare the old rate with 0.5/L and 1/L. Use tolerance 1e-12, patience 20 and a 100,000-step cap. Only compare models that pass the training-optimum check.

Selected degree 3, rate 0.23337059, 2848 updates. Validation R2 0.70375, RMSE 98.50538. This is a modest improvement, not a solution to all residual structure. Save degree, expansion basis, scaling and actual weights for Phase Three.

Source: supplied CSV files and executed bike_sharing_demand.ipynb

Phase Three | Bias, variance and time

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

Vary degrees 1-5 around the Phase Two degree-3 anchor, retaining the same weather-power family. All fits pass the training-optimum check so unfinished optimization is not mistaken for model bias.

Make one chronological split at 2012-08-05, keeping dates intact: 8703 earlier training and 2183 later validation rows. Fit all transformations on the earlier period only; do not carry future-informed random-split weights into chronological fitting.

At degree 3, random validation RMSE is 98.5054, versus 130.3631 on later dates; R2 changes from 0.7038 to 0.6409. The later-period estimate is more relevant for future prediction, but a single cutoff is not a confidence interval.

Diagnosis: the random-split model fits reasonably within this feature family. Degree 5 shows a small overfitting sign. Performance is weaker on later dates. Demand patterns can change over time, so this gap alone does not prove that overfitting is the cause.

Carry degree 4 forward because it minimizes required seeded-validation RMSE; chronology instead favors degree 3. Disclose that conflict and the small random gain. This motivates weak/moderate regularization comparisons, not an assumption that strong shrinkage fixes time shift.

Degree	random train	random validation	chrono train	chrono validation
1	100.2631	99.4655	91.8688	131.0797
2	99.8886	99.1821	91.4592	131.0799
3	99.1686	98.5054	90.7208	130.3631
4	98.9808	98.3458	90.3616	131.0821
5	98.9299	98.3803	90.3173	131.0440

Source: supplied CSV files and executed bike_sharing_demand.ipynb

Phase Four | Regularization and selection

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

Implement Lasso (L1), Ridge (L2) and Elastic Net using half-MSE plus a penalty; do not penalize the intercept. Soft thresholding creates exact zeros for L1. The KKT residual checks whether the weights minimize the penalized loss. Use tolerance 1e-7*(1+max(abs(b))) and a 100,000-step cap. All 35 candidates use the same rows and scaling.

Search lambda 0.0001 through 100; Elastic Net ratios 0.2/0.5/0.8. Lasso selects lambda 0.01; Ridge lambda 0.0001; Elastic Net lambda 0.0001, ratio 0.8. Lasso's validation RMSE 98.34383 improves unpenalized 98.34580 by only 0.00197: a negligible accuracy gain, but 57/62 terms remain for Phase Five.

Refit after removing each original input's derived terms at fixed selected settings. Useful means removal cost >0.1 rentals/hour; otherwise redundant if feature reconstruction R2>=0.90, uninformative otherwise. These are conditional practical labels, not significance tests. The index row instead shows add-feature gain. Date means incremental day-of-month; target cnt is never an input.

L1 feature lists and column-removal tests answer different questions. Correlated features can replace each other after refitting; do not delete every redundant group simultaneously. Retain the exact Lasso subset. No evidence here establishes that the time gap is repaired.

Input	Verdict	removal RMSE delta	reconstruct R2
dteday	uninformative	0.08919	0.02710
season	redundant	-0.00179	1.00000
yr	useful	9.43777	0.04640
mnth	useful	1.19885	0.35750
hr	useful	45.72496	0.01519
holiday	redundant	-0.00521	1.00000
weekday	useful	0.11684	0.14247
workingday	redundant	-0.00143	1.00000
weathersit	useful	1.17817	0.15457
temp	useful	1.08211	0.61497
atemp	redundant	0.04207	0.97150
hum	useful	0.39033	0.29536
windspeed	useful	0.11358	0.10468
instant	redundant	0.00912	0.99999

Source: supplied CSV files and executed bike_sharing_demand.ipynb

Phase Five | Contextual high-demand classifier

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

Define high demand as count above the training 75th percentile for the same hour and working-day status. This relative service-pressure flag avoids labeling nearly every busy hour positive and nearly every night negative. It is not an absolute station-capacity rule. Use a 30-row minimum group, with training hour/global fallbacks; strict greater-than handles ties.

Training positive rate 24.633%; validation 25.574%. No validation quantiles are fitted. The classifier uses exactly the 57 Lasso survivors from Phase Four, with unchanged transformations and rows.

Fit logistic regression with stable probability/loss formulas and damped Newton updates. Check the gradient by finite differences, require loss to decrease, and stop below gradient 1e-7 with a 100-step cap. Test L2 strengths 0/0.0001/0.001/0.01/0.1 on the same selected inputs.

Select L2 lambda 0 by validation ROC-AUC and probability threshold 0.44 by validation F1. These are separate from the fixed count cutoffs. Selected validation accuracy 0.8788, F1 0.7596, ROC-AUC 0.9160, precision 0.7708, recall 0.7487.

At probability 0.5, accuracy/F1 are 0.8811/0.7531. The training-majority baseline has accuracy 0.7443 and F1 0.0000. AUC describes ranking; F1/precision/recall describe the chosen alert tradeoff.

Outcome: use the measured comparison, not accuracy alone, to assess usefulness. Penalty and threshold selection reuse validation, making reported scores tuned estimates. These classifier results do not prove good performance on future dates.

Pipeline retrospective and submission

Bike-sharing demand | IDs: 14004179, 16003110, 16003585 | seed 29578

The chain is numeric and inspectable: Phase One weights -> Phase Two warm-started powers -> Phase Three diagnosis/degree -> Phase Four penalties/L1 mask -> Phase Five classifier. Each artifact stores ordered features, split IDs, transformations and upstream hashes.

The regression baseline improves modestly with selective weather curvature. Degree 4 gives the best tested seeded estimate; the chronological deterioration is much larger than the later regularization gain. L1 contributes mainly sparsity. Classification addresses a different relative-demand question, so its scores cannot be compared directly with regression R2/RMSE.

Final count predictions use Lasso, lambda 0.01, refitted on all 10886 labeled rows using already-selected settings and existing transformations. The brief permits this final refit. Its coefficients are saved separately; no refit-on-all-data validation score is reported.

The root sample_submission.csv contains 574 test rows in original instant order, with columns instant,cnt. Clip impossible negative count estimates at zero without rounding; do not submit classifier probabilities as counts. The original data/sample_submission.csv template remains unchanged.

For transparency, pre-refit raw validation RMSE is 98.34383; zero-clipped RMSE is 95.92731. This zero clipping does not rewrite earlier raw-model comparisons.

Limitations: one repeatedly reused validation split; a single chronological cutoff; relative rather than absolute high-demand labels; additive weather powers omit interactions; sparse coefficients need not imply causal or universal feature importance. Hidden test data are never used for tuning. Review the notebook and README to explain or rerun every choice.

Deliverables: bike_sharing_demand.ipynb (all phases with Expectation/Outcome), this six-page project_report.pdf, sample_submission.csv, and artifacts/phase_one.json through phase_five.json. All figures and numbers come from the supplied data and this executed pipeline.

Source: supplied CSV files and executed bike_sharing_demand.ipynb